ANALYSIS REPORT

Wave 3.2 Harmonic-Offset Probe Campaign Results

Wave 3.2 Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Wave 3.2` harmonic-offset probe campaign. The campaign goal was to keep the clean `Wave 3.1` non-harmonic residual-offset baseline alive, then test a harmonic-forced residual-offset branch on the same three required deployment surfaces: `global`, `Fw`, and `Bw`.

The first execution completed the three clean baseline runs and failed the three harmonic-offset runs before training started because the campaign runner did not yet route the new `harmonic_residual_offset_probe` model type. The runner registration has now been fixed, the three failed harmonic configs were repaired through a dedicated rerun, and the effective closeout state is six trained candidates with zero unresolved failed runs.

The scalar leaderboard ranks `Fw` first by `test_mae`, but that ranking remains diagnostic only. Wave 3.2 keeps three parallel branch outcomes for future curve-first verification and deployment analysis: one `global`, one `Fw`, and one `Bw`. The clean and harmonic branches are also both retained because the clean branch is the no-harmonic baseline for later multi-head, composite-loss, and new-index work.

Campaign Artifacts

Artifact	Path
Original campaign output	`output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04`
Repair campaign output	`output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05`
Original leaderboard	`output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04/campaign_leaderboard.yaml`
Repair leaderboard	`output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05/campaign_leaderboard.yaml`
Original execution report	`output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04/campaign_execution_report.md`
Repair execution report	`output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/track_2/2026-06-04-22-57-04_track2f_bis_harmonic_offset_probe_campaign_plan_report.md`
Campaign technical document	`doc/technical/2026-06/2026-06-04/2026-06-04-21-14-52_track2f_bis_harmonic_offset_probe.md`
Runner-fix technical document	`doc/technical/2026-06/2026-06-05/2026-06-05-15-56-59_track2f_bis_campaign_runner_model_type_fix.md`

Execution Summary

Field	Value
Original campaign name	`track2f_bis_harmonic_offset_probe_campaign_2026_06_04`
Repair campaign name	`track2f_bis_harmonic_offset_probe_repair_2026_06_05`
Original completed runs	3
Original failed runs	3
Repair completed runs	3
Repair failed runs	0
Effective completed candidates	6
Unresolved failed candidates	0
Clean model type	`sequential_residual_offset_probe`
Harmonic model type	`harmonic_residual_offset_probe`
Tested surfaces	`global` , `Fw` , `Bw`
Runner-level scalar first entry	`te_track2f_bis_harmonic_residual_offset_fw`

Directional Branch Results

Surface	Role	Run	Family	Test MAE	Test RMSE	Val MAE	Params
global	clean baseline	`te_track2f_bis_clean_residual_offset_global`	`track2f_bis_clean_sequential_residual_offset_global`	0.003528	0.004010	0.003717	92,802
global	harmonic offset	`te_track2f_bis_harmonic_residual_offset_global`	`track2f_bis_harmonic_residual_offset_global`	0.003538	0.003932	0.003659	85,747
Fw	clean baseline	`te_track2f_bis_clean_residual_offset_fw`	`track2f_bis_clean_sequential_residual_offset_fw`	0.003446	0.003972	0.003474	92,802
Fw	harmonic offset	`te_track2f_bis_harmonic_residual_offset_fw`	`track2f_bis_harmonic_residual_offset_fw`	0.002862	0.003334	0.002941	85,747
Bw	clean baseline	`te_track2f_bis_clean_residual_offset_bw`	`track2f_bis_clean_sequential_residual_offset_bw`	0.003540	0.004203	0.003820	92,802
Bw	harmonic offset	`te_track2f_bis_harmonic_residual_offset_bw`	`track2f_bis_harmonic_residual_offset_bw`	0.003336	0.003935	0.003555	85,747

These six rows are the closeout result. The harmonic `Fw` row is the runner-level scalar first entry, but it does not replace the required `global` or `Bw` branch candidates.

Clean Versus Harmonic Delta

Surface	Clean MAE	Harmonic MAE	MAE Gain [%]	Clean RMSE	Harmonic RMSE	RMSE Gain
global	0.003528	0.003538	-0.28	0.004010	0.003932	1.94
Fw	0.003446	0.002862	16.96	0.003972	0.003334	16.91
Bw	0.003540	0.003336	5.75	0.004203	0.003935	6.38

The scalar result is direction-dependent. Harmonic forcing improves `Fw` clearly, improves `Bw` moderately, and leaves `global` almost unchanged on `test_mae` while still reducing `test_rmse` . This is enough to keep the harmonic-offset branch alive, but it is not enough to make a deployment decision before the official curve-first TE Curve Verification refresh.

Failure Repair

The first campaign failure was not a model-convergence result. The three harmonic runs stopped immediately because `scripts/training/run_training_campaign.py` did not include `harmonic_residual_offset_probe` in the campaign model-type dispatch map.

The fix adds `harmonic_residual_offset_probe` to the campaign runner and routes it through the existing feedforward training entry point. A one-batch setup validation was then run on the failed global queue config, followed by a repair campaign over the three failed harmonic configs. The repair campaign completed all three runs and produced the expected leaderboard, best-run, family registry, and program-registry artifacts.

The launcher wrapper was also hardened for a recurring operator issue. When the PowerShell launcher is already running inside the requested Conda environment, it now calls `python` directly instead of nesting another `conda run`. This prevents the misleading terminal-level `conda run ... failed` message seen in the previous Wave 3.1 and Wave 3.2 operator runs.

Scalar Leaderboard

Rank	Surface	Run	Family	Test MAE	Test RMSE	Val MAE	Params
1	Fw	`te_track2f_bis_harmonic_residual_offset_fw`	`track2f_bis_harmonic_residual_offset_fw`	0.002862	0.003334	0.002941	85,747
2	Bw	`te_track2f_bis_harmonic_residual_offset_bw`	`track2f_bis_harmonic_residual_offset_bw`	0.003336	0.003935	0.003555	85,747
3	Fw	`te_track2f_bis_clean_residual_offset_fw`	`track2f_bis_clean_sequential_residual_offset_fw`	0.003446	0.003972	0.003474	92,802
4	global	`te_track2f_bis_clean_residual_offset_global`	`track2f_bis_clean_sequential_residual_offset_global`	0.003528	0.004010	0.003717	92,802
5	global	`te_track2f_bis_harmonic_residual_offset_global`	`track2f_bis_harmonic_residual_offset_global`	0.003538	0.003932	0.003659	85,747
6	Bw	`te_track2f_bis_clean_residual_offset_bw`	`track2f_bis_clean_sequential_residual_offset_bw`	0.003540	0.004203	0.003820	92,802

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
`track2f_bis_clean_sequential_residual_offset_global`	`te_track2f_bis_clean_residual_offset_global`	0.003528	Clean global baseline retained
`track2f_bis_clean_sequential_residual_offset_fw`	`te_track2f_bis_clean_residual_offset_fw`	0.003446	Clean forward baseline retained
`track2f_bis_clean_sequential_residual_offset_bw`	`te_track2f_bis_clean_residual_offset_bw`	0.003540	Clean backward baseline retained
`track2f_bis_harmonic_residual_offset_global`	`te_track2f_bis_harmonic_residual_offset_global`	0.003538	Harmonic global branch trained
`track2f_bis_harmonic_residual_offset_fw`	`te_track2f_bis_harmonic_residual_offset_fw`	0.002862	Harmonic forward branch trained
`track2f_bis_harmonic_residual_offset_bw`	`te_track2f_bis_harmonic_residual_offset_bw`	0.003336	Harmonic backward branch trained

The campaign runner refreshed the family registries for all six branch families and updated the program registry. The program-level scalar best did not move because the existing Wave 2.2 periodic `GRU` backward-only model is still stronger on scalar `test_mae`.

TE Curve Verification Pipeline Boundary

Official TE curve-first verification was not run as part of this closeout. Under campaign governance, that remains a separate operator-approved workflow after the final campaign-results report and PDF are complete.

The next verification package must evaluate the clean and harmonic Wave 3.2 branches on `global`, `Fw`, and `Bw` in parallel. The intended decision is not "one best model overall" at this stage. The intended decision is whether the harmonic-offset intervention improves curve following for each deployment surface while preserving the clean non-harmonic baseline as a control.

Closeout Decision

Wave 3.2 is complete from an execution standpoint after repair: all six planned candidates now have successful training artifacts, no unresolved failed run remains, and the active campaign state can be cleared.

From a modeling standpoint, Wave 3.2 is a useful intervention branch but not yet a deployment promotion. The scalar metrics support carrying the harmonic `Fw` and `Bw` branches forward, while the `global` branch needs curve-first verification before any stronger conclusion.

Recommended Follow-Up

1. Keep six Wave 3.2 candidates available: clean and harmonic variants for `global`, `Fw`, and `Bw`.
2. Do not collapse the branch set to the scalar-first harmonic `Fw` candidate.
3. Run the separate official TE Curve Verification refresh once the closeout PDF is accepted.
4. Inspect the curve overlays before deciding whether harmonic-offset should become the next training family or remain a comparison branch.